Chapter 7 — Waves and Vibrations

Complete Notes · PECTAA Physics 11 · Board-style longs + conceptual shorts

Topic 1 — Waves and the principle of superposition

Q.1 Explain the basic quantities of a wave and the principle of superposition.

A wave transfers energy without transferring matter. In a transverse wave the particles vibrate perpendicular to the direction of travel, as in light and waves on a string; in a longitudinal wave they vibrate parallel to the direction of travel, as in sound. The main quantities are wavelength λ, frequency f, period T and speed v, related by $v = f\lambda$. The intensity of a wave is the power per unit area and is proportional to the square of the amplitude.

The principle of superposition states that when two or more waves overlap, the resultant displacement at any point is the vector sum of the individual displacements. This principle explains interference, beats, stationary waves and noise-cancelling headphones.

Important Conceptual Short Questions — Topic 1

1. Why does a wave transfer energy but not matter?

The particles of the medium only oscillate about their mean positions. The disturbance and its energy pass from particle to particle, but there is no net transport of the medium itself.

2. How does superposition explain noise-cancelling headphones?

The headphones produce a sound wave with the same amplitude but opposite phase to the unwanted noise. The two waves superpose destructively, so the resultant displacement and the heard sound are reduced.

3. Why is intensity proportional to the square of amplitude?

The energy of a vibrating particle is proportional to the square of its amplitude. Since intensity is energy flowing per unit area per second, it also varies as amplitude squared.

Topic 2 — Interference of waves

Q.2 Explain constructive and destructive interference and the conditions for each.

Interference is the superposition of two coherent waves of the same frequency. Constructive interference occurs when the waves arrive in phase, so the resultant amplitude is maximum. The path difference must be an integral number of wavelengths: $\Delta x = n\lambda$, where $n = 0, 1, 2, \ldots$ Destructive interference occurs when the waves arrive out of phase by half a wavelength, so the resultant amplitude is minimum. The path difference is $\Delta x = (2n + 1)\lambda/2$.

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Coherent sources have a constant phase difference and the same frequency. In a ripple tank or with two loudspeakers, regions of loud sound or large ripples alternate with quiet or calm regions, showing the interference pattern.

Important Conceptual Short Questions — Topic 2

1. Why must the sources be coherent to obtain a steady interference pattern?

A steady pattern requires a constant phase difference at each point. If the phase difference changes randomly, the positions of maxima and minima shift rapidly and no fixed pattern is seen.

2. How does path difference determine whether interference is constructive or destructive?

A path difference of $n\lambda$ brings the waves back in phase, giving constructive interference. A path difference of $(2n + 1)\lambda/2$ brings them out of phase, giving destructive interference.

3. Why is the sound loud at some points and faint at others between two speakers?

At points where the path difference is $n\lambda$ the waves reinforce, producing loud sound. At points where it is $(2n + 1)\lambda/2$ they cancel, producing faint or no sound.

Topic 3 — Stationary waves on a stretched string

Q.3 Explain the formation of stationary waves and derive the frequencies of a stretched string.

A stationary wave is formed when two identical waves travelling in opposite directions superpose. It has nodes, where the displacement is always zero, and antinodes, where the displacement is maximum. Adjacent nodes are $\lambda/2$ apart, and a node and the next antinode are $\lambda/4$ apart. No energy flows past a node.

For a string of length ℓ fixed at both ends, the ends must be nodes. The allowed wavelengths are $\lambda_n = 2\ell/n$, so the frequencies are $f_n = nv/2\ell$, where $n = 1, 2, 3, \ldots$ and $v = \sqrt{(F/\mu)}$, with F the tension and μ the mass per unit length. The fundamental is $f_1 = v/2\ell$, and the higher frequencies are harmonics.

Important Conceptual Short Questions — Topic 3

1. Why is no energy transmitted past a node in a stationary wave?

The medium at a node remains at rest, so no work is done across it. Energy is confined between nodes and alternates between kinetic and potential forms.

2. How does increasing the tension of a string change its frequency?

The wave speed is $v = \sqrt{(F/\mu)}$, so increasing tension increases the speed. Since $f = nv/2\ell$, the frequency of each harmonic increases.

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3. Why are the ends of a stretched string nodes?

The ends are fixed and cannot move. A stationary wave must therefore have zero displacement at the ends, which is the definition of a node.

Topic 4 — Stationary waves in air columns

Q.4 Compare stationary waves in an open pipe and a pipe closed at one end.

In an open pipe both ends are antinodes. The fundamental wavelength is $\lambda_1 = 2\ell$, so $f_1 = v/2\ell$, and all harmonics $f_n = nv/2\ell$ are present. In a pipe closed at one end, the closed end is a node and the open end is an antinode. The fundamental wavelength is $\lambda_1 = 4\ell$, so $f_1 = v/4\ell$, and only odd harmonics $f_n = nv/4\ell$ (n = 1, 3, 5, ...) are present.

Feature	Open pipe	Closed pipe
Ends	Antinode at both ends	Node at closed end, antinode at open end
Fundamental	$f_1 = v/2\ell$	$f_1 = v/4\ell$
Harmonics	All harmonics	Odd harmonics only
Quality	Richer in harmonics	Fewer harmonics

Textbook check (p. 146): the frequency $3f_1$ of a closed pipe is called the second harmonic. It is usually called the third harmonic, or the first overtone.

Important Conceptual Short Questions — Topic 4

1. Why does a pipe closed at one end sound lower than an open pipe of the same length?

Its fundamental frequency is $v/4\ell$, while that of an open pipe is $v/2\ell$. The closed pipe therefore has half the fundamental frequency and sounds lower.

2. Why are only odd harmonics present in a closed pipe?

The closed end must be a node and the open end an antinode. This boundary condition allows only wavelengths $\lambda = 4\ell/n$ with n odd, so only odd harmonics fit.

3. How does a microwave experiment demonstrate stationary waves?

Microwaves reflected from a metal plate superpose with the incident waves to form nodes and antinodes. A detector moved between the source and plate shows alternating maxima and minima, confirming the stationary wave pattern.

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Topic 5 — Diffraction and beats

Q.5 Explain diffraction of waves and the formation of beats.

Diffraction is the spreading of waves around obstacles or through gaps. It is most noticeable when the size of the gap or obstacle is comparable with the wavelength. In a ripple tank, water waves spread out after passing through a narrow gap, showing that waves do not travel only in straight lines.

Beats are the periodic rise and fall of loudness when two sound waves of slightly different frequencies superpose. The beat frequency equals the difference of the two frequencies: $f_{beat} = |f_1 - f_2|$. Beats become difficult to recognise when the difference exceeds about 10 Hz.

Textbook check (p. 152): the text says the frequency difference is 4 Hz, but forks of 32 Hz and 30 Hz differ by 2 Hz, as the calculation below it shows.

Important Conceptual Short Questions — Topic 5

1. Why is diffraction more noticeable when the gap is comparable with the wavelength?

When the gap is much larger than the wavelength, the wave passes almost straight. When the gap is comparable with the wavelength, the wave spreads strongly into the region beyond the gap.

2. How are beats produced by two tuning forks?

The two waves alternately reinforce and cancel as their phase difference changes. This produces a slow rise and fall of loudness whose frequency equals the difference of the fork frequencies.

3. Why do beats become hard to recognise above about 10 Hz?

The loudness variations become too rapid for the ear to follow as separate beats, so they are heard as a rough or continuous tone instead.

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Topic 6 — Doppler effect and its applications

Q.6 Explain the Doppler effect for sound and derive the observed frequency when the source moves.

The Doppler effect is the apparent change in frequency due to relative motion between source and observer. When a source moves toward a stationary listener with speed u_s, the wavefronts are crowded together and the observed frequency increases: $f' = [v/(v - u_s)]f$. When the source moves away, the wavefronts are spread out and $f' = [v/(v + u_s)]f$, so the frequency decreases.

Applications include radar speed traps, sonar, astronomy and medical heart scans. A receding star shows a red shift because its light is shifted to lower frequency, while an approaching source shows a blue shift.

Textbook check (p. 156): the scientist is named John Doppler. The correct name is Christian Doppler.

Important Conceptual Short Questions — Topic 6

1. Why does the pitch of a whistle rise as a train approaches?

The moving source crowds the wavefronts in front of it, decreasing the wavelength. Since the speed of sound is unchanged, the observed frequency increases and the pitch rises.

2. How does the Doppler effect help astronomers measure the motion of stars?

Light from a receding star is shifted to lower frequency, called red shift, and light from an approaching star is shifted to higher frequency. The size of the shift gives the speed along the line of sight.

3. Why does a moving source change the observed frequency but not the speed of sound?

The speed of sound depends on the medium, not on the source. Motion of the source changes the spacing of the wavefronts, so the wavelength and observed frequency change while the wave speed stays the same.

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